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Statement of work.

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Main Scenario and Background		
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To the best of my ability, what I have stated is true and follows the guidance and instructions of the group work assessment for this has been submitted with.

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Assessing Cyber-Physical Risks in Metro Stations: A Layered Approach to Enhancing Security and Resilience

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1.Introduction

The rapid advancement of technology and the increasing reliance on interconnected systems have transformed the way metro stations operate. While these advancements have brought about improved efficiency, passenger experience, and safety, they have also introduced new vulnerabilities and risks that must be addressed. This risk analysis report aims to provide a comprehensive assessment of the cyber-physical risks associated with metro stations, focusing on the complex interplay between Information Technology (IT), Operational Technology (OT), and safety systems. By examining these risks through a layered view of the hybrid environment, we can better understand the potential impact of cyberattacks on metro station operations, passenger safety, and overall security.

The specific objectives of this research are as follows:

1. To identify and categorize the various cyber-physical risks present in metro stations, considering the unique characteristics of IT, OT, and safety systems.
2. To analyse the potential consequences of successful cyberattacks on metro stations, including financial losses, operational disruptions, compromised passenger safety, and reputational damage.
3. To develop a comprehensive set of recommendations for mitigating the identified risks, considering industry best practices, relevant standards, and case studies of successful cybersecurity implementations in similar environments.

By addressing these objectives, this report aims to contribute to the ongoing efforts to safeguard metro stations against evolving cyber threats and ensure the continuity of safe and reliable transportation services for millions of passengers worldwide. The report will begin by providing a detailed background on the current state of metro station technology and the growing concerns surrounding cyber threats. It will then delve into the scenario analysis, breaking down the hybrid environment into its constituent layers and identifying the specific risks associated with each layer. The report will also discuss the potential consequences of successful cyberattacks, ranging from financial losses and operational disruptions to compromised passenger safety and loss of life.

Finally, the report will present a set of comprehensive recommendations for mitigating these risks, considering the unique challenges posed by the convergence of IT, OT, and safety systems. These recommendations will be based on industry best practices, relevant standards, and case studies of successful cybersecurity implementations in similar environments. By providing a thorough and evidence-based analysis of the cyber-physical risks in metro stations, this report aims to equip IT specialists and decision-makers with the knowledge and insights necessary to develop and implement effective risk mitigation strategies. The goal is to ensure the safety, security, and reliability of metro station operations in the face of evolving cyber threats.

2.Main scenario and background

Metro stations are essential components of urban transportation infrastructure, serving millions of passengers daily. The efficient operation of these stations relies on a complex network of interconnected systems, including train control systems, signalling systems, power systems, ticketing systems, passenger information systems, and security systems. The increasing digitization and automation of these systems have brought about significant benefits, such as improved efficiency, reduced costs, and enhanced passenger experience.

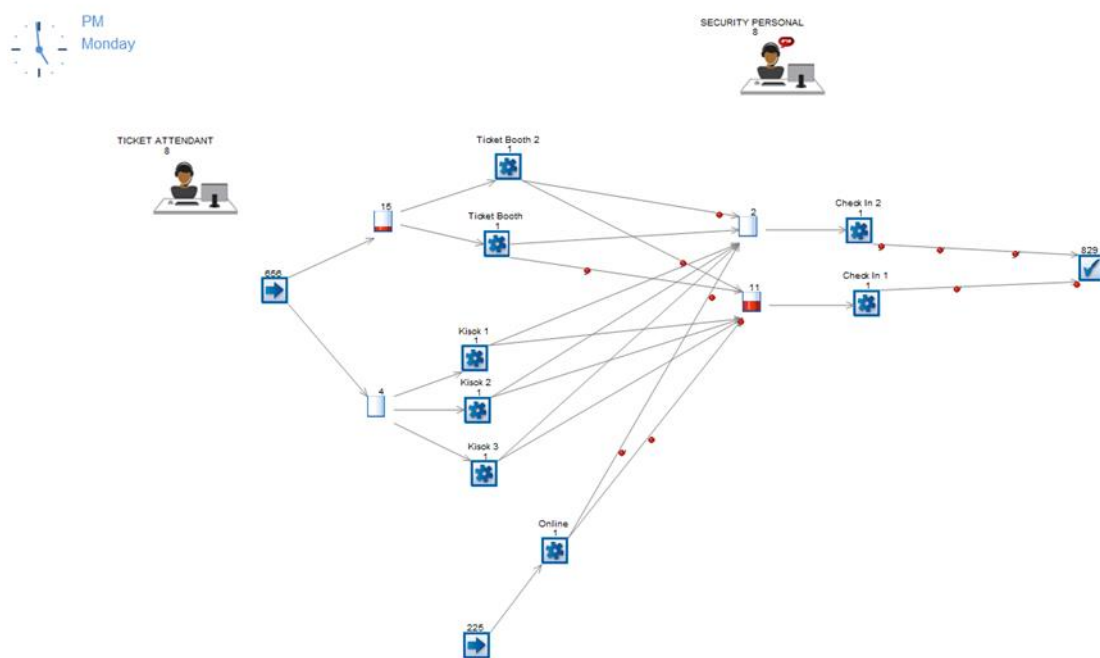


Figure 1 - Simulation of metro station.

However, this growing reliance on technology has also introduced new vulnerabilities and risks. As IT and OT systems become more interconnected, the attack surface expands, creating opportunities for malicious actors to exploit weaknesses and cause significant harm. (Madnick, n.d.) This convergence of IT and OT systems, known as the Industrial Internet of Things (IIoT), has been a growing concern for critical infrastructure sectors, including transportation.



Figure 2 - Metro station kiosk after a cyber-attack.

Real-world incidents have highlighted the potential consequences of cyberattacks on metro stations. In 2016, the San Francisco Municipal Transportation Agency (SFMTA) fell victim to a ransomware attack that affected its ticketing systems, forcing the agency to provide free rides to passengers for two days. The attack resulted in financial losses and operational disruptions, underscoring the need for robust cybersecurity measures. More recently, in 2021, the Metropolitan Transportation Authority (MTA) in New York City suffered a cyberattack that compromised its computer systems, forcing the agency to temporarily shut down its online services and email systems. While the attack did not affect train operations or passenger safety, it highlighted the potential for more severe consequences if critical systems were compromised. (The Guardian, n.d.)

These incidents, along with the growing frequency and sophistication of cyberattacks targeting critical infrastructure, underscore the urgent need for a comprehensive approach to cybersecurity in metro stations. This approach must consider the unique challenges posed by the convergence of IT, OT, and safety systems, as well as the potential impact on passengers, staff, and the wider community.

By examining the cyber-physical risks in metro stations through a layered view of the hybrid environment, this report aims to provide a foundation for developing effective risk mitigation strategies that address the complex interplay between technology, operations, and safety. These incidents, along with the growing frequency and sophistication of cyberattacks targeting critical infrastructure, underscore the urgent need for a comprehensive approach to cybersecurity in metro stations.

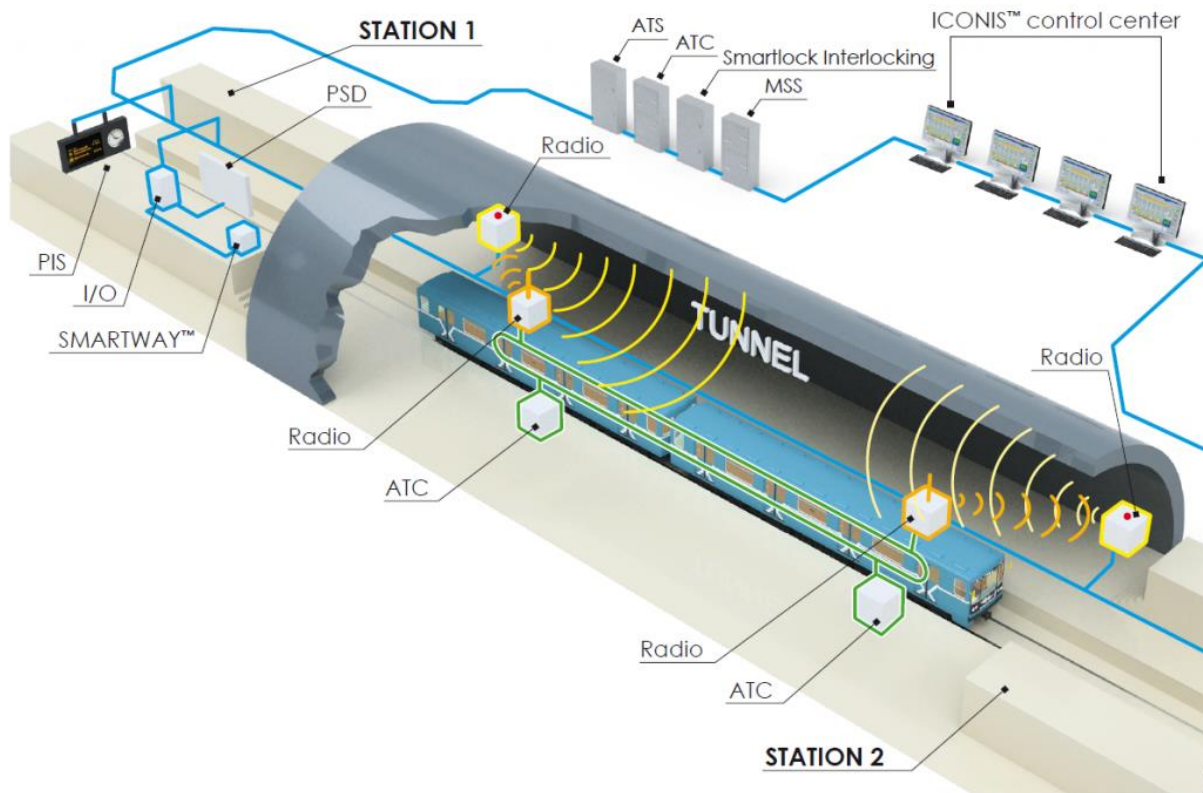


Figure 3 : Metro station overview

The layered view of the hybrid environment in metro stations consists of the following layers:

1. **Physical Layer:** This layer encompasses the physical infrastructure and assets of the metro station, including tracks, trains, stations, platforms, and passenger areas. According to the International Association of Public Transport (UITP), there are over 200 metro systems worldwide, with a total network length of more than 17,000 km and serving over 582 billion passengers on 2019 alone. The physical layout of a typical metro station consists of multiple levels, including the entrance level, concourse level, and platform level. The entrance level provides access to the station and may include ticketing facilities, turnstiles, and passenger information displays. The concourse level serves as a distribution area, connecting the entrance level to the platforms and often housing amenities such as shops, restrooms, and seating areas. The platform level is where passengers board and disembark trains, with platforms designed to accommodate the specific train types and passenger volumes of the station. (TRANSPORT, 2022)
2. **Operational Technology (OT) Layer:** The OT layer consists of the systems and devices that control and monitor the physical processes in the metro station, such as train control systems, signalling systems, power systems, and environmental control systems. In the image, the ATC system is a prime example of an OT system, responsible for controlling the movement and speed of the trains. Other OT systems include the

platform screen doors, which ensure passenger safety, and the radio communication system that enables communication between the trains and the control centre. These systems are often legacy and proprietary, with limited built-in security features. A survey by the Ponemon Institute found that 65% of OT organizations experienced at least one cybersecurity incident in the past 12 months. (Ponemon INSTITUTE, n.d.)

3. **Information Technology (IT) Layer:** The IT layer includes the systems and applications that support the business processes and passenger services in the metro station, such as ticketing systems, passenger information systems, surveillance systems, and communication networks. A study by the PRECEDENCE RESEARCH predicts that worldwide spending on IoT in transportation will reach \$473.69 billion by 2030. (RESEARCH, n.d.) Ticketing systems facilitate the sale and validation of fare media, such as smart cards, tokens, and mobile tickets, and often integrate with fare gates and turnstiles to control access to the station. Passenger information systems provide real-time updates on train arrivals, departures, and service disruptions through displays, announcements, and mobile applications. Surveillance systems, including CCTV cameras and access control systems, monitor and secure the station environment, while communication networks enable data exchange and coordination between various systems and personnel.
4. **Communication Layer:** This layer encompasses the networks and protocols that enable communication and data exchange between the OT and IT systems, as well as with external entities such as remote-control centres and third-party service providers. The communication layer often relies on wireless technologies, such as Wi-Fi, 4G/5G, and Bluetooth, which can introduce additional security risks. Metro stations typically employ a combination of wired and wireless networks to support the various systems and services. Wired networks, such as Ethernet and fibre optic cables, provide high-bandwidth, low-latency connections for critical systems like train control and signalling. Wireless networks offer flexibility and scalability for passenger-facing services, such as Wi-Fi access and mobile ticketing, but require careful security considerations to prevent unauthorized access and data interception. (Alshehri, n.d.)
5. **Human Layer:** The human layer consists of the personnel involved in the operation and maintenance of the metro station, including employees, contractors, and passengers. According to a report by Kaspersky, 52% of industrial control system (ICS) incidents in transportation were caused by human error or unintentional actions. Metro station staff, such as train operators, station attendants, and maintenance crews, play a crucial role in ensuring the safe and efficient operation of the station. They must be trained in the proper use of the various systems and equipment, as well as in emergency response procedures. Contractors, including vendors and service providers, may have access to sensitive systems and data, necessitating strict access controls and security vetting processes. Passengers, while not directly involved in station operations, can impact security through their behaviour and interactions with the station environment,

such as by reporting suspicious activities or falling victim to social engineering attacks. (Kaspersky ICS CERT, n.d.)

Cyber-Physical Risks:

1. IT Risks:

- Ransomware attacks on ticketing systems, leading to financial losses, operational disruptions, and reputational damage.
- Data breaches involving passenger information and payment details, resulting in identity theft, financial fraud, and legal liabilities.
- Malware infections spreading across the IT network, compromising multiple systems, and causing widespread disruption.
- Phishing attacks targeting employees, potentially providing attackers with initial access to the network and enabling further compromise.

2. OT Risks:

- Unauthorized access to train control systems, potentially causing collisions, derailments, or other safety incidents.
- Manipulation of signalling systems, leading to unsafe train movements, delays, and service disruptions.
- Disruption of power systems, causing train stoppages, station blackouts, and safety hazards.
- Tampering with surveillance systems, compromising station security, and hindering incident response.

3. Safety Risks:

- Compromised train control systems leading to accidents, injuries, or loss of life. Attackers may manipulate speed limits, override safety protocols, or disable emergency brakes, resulting in catastrophic failures and human casualties.
- Failure of emergency communication systems during crisis situations, delaying or preventing the dispatch of help and evacuation of passengers. Attackers may disrupt the communication channels, jam the signals, or overwhelm the systems with traffic.
- Inadequate incident response due to compromised security systems and lack of situational awareness. Attackers may disable alarms, suppress alerts, or provide false information to the control centre, impairing the ability to detect and respond to security incidents.

4. Hybrid Risks:

- Cascading effects of cyberattacks across IT and OT systems, amplifying the impact on operations and safety. A successful attack on one layer may propagate to other layers, causing a chain reaction of failures and disruptions.
- Exploitation of the communication layer to pivot between IT and OT networks, expanding the attack surface and enabling the compromise of critical systems. Attackers may use the communication channels as a bridge to infiltrate isolated or air-gapped networks.
- Inadequate segmentation and access controls between IT and OT networks, facilitating the spread of attacks and increasing the potential damage. Lack of proper network zoning, firewalls, and authentication mechanisms may allow attackers to move laterally and escalate their privileges.

By analysing the cyber-physical risks across the different layers of the hybrid environment, we can gain a deeper understanding of the potential impact of cyberattacks on metro station operations, safety, and security. This knowledge is crucial for developing effective risk mitigation strategies that consider the unique characteristics and interdependencies of each layer.

3. Second Layer to Main Scenario

In recent years, Communication Based Train Control System (CBTC) has emerged as the leading automatic train control system (ATC) for metro systems worldwide. With its widespread adoption, predominantly employing Wi-Fi based train-to-track communication, and reliance on Ethernet-based computer networks for communication links between operations control centres, interlocking systems, and CBTC zone controllers, CBTC technology has become integral to modern railway signalling systems. However, this technological advancement also brings along cybersecurity concerns common to the Information Technology (IT) industry, such as unauthorized access and control by external parties. The Operational Technology (OT) layer is a critical component of the metro station's hybrid environment, as it encompasses the systems and devices that directly control and monitor the physical processes essential for the safe and efficient operation of trains and the station itself. This layer includes a wide range of systems, such as:

- **Train Control Systems:** These systems are responsible for managing the movement of trains, ensuring safe separation between them, and controlling their speed and acceleration. They rely on complex algorithms, sensors, and communication protocols to maintain optimal performance and prevent collisions.

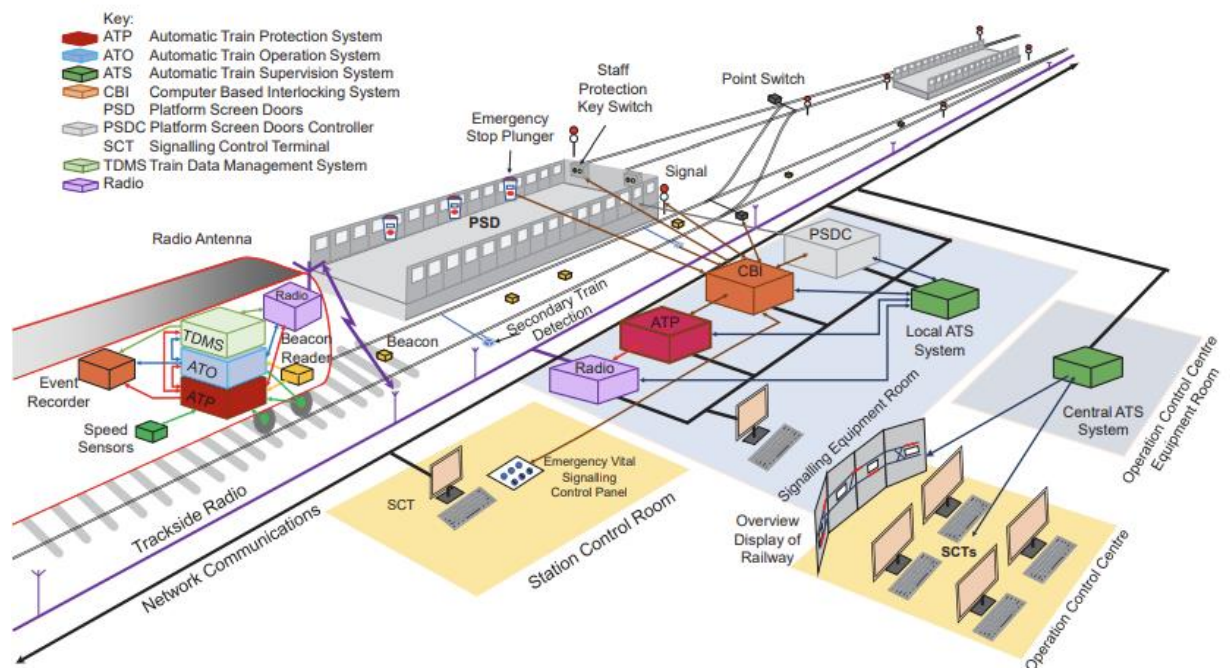


Figure 4 - Train control systems overview.

- **Signalling Systems:** Signalling systems work in conjunction with train control systems to regulate traffic flow, manage route settings, and ensure the safe movement of trains. They use a combination of trackside and on-board equipment to convey information about the status of the track ahead and the position of other trains.

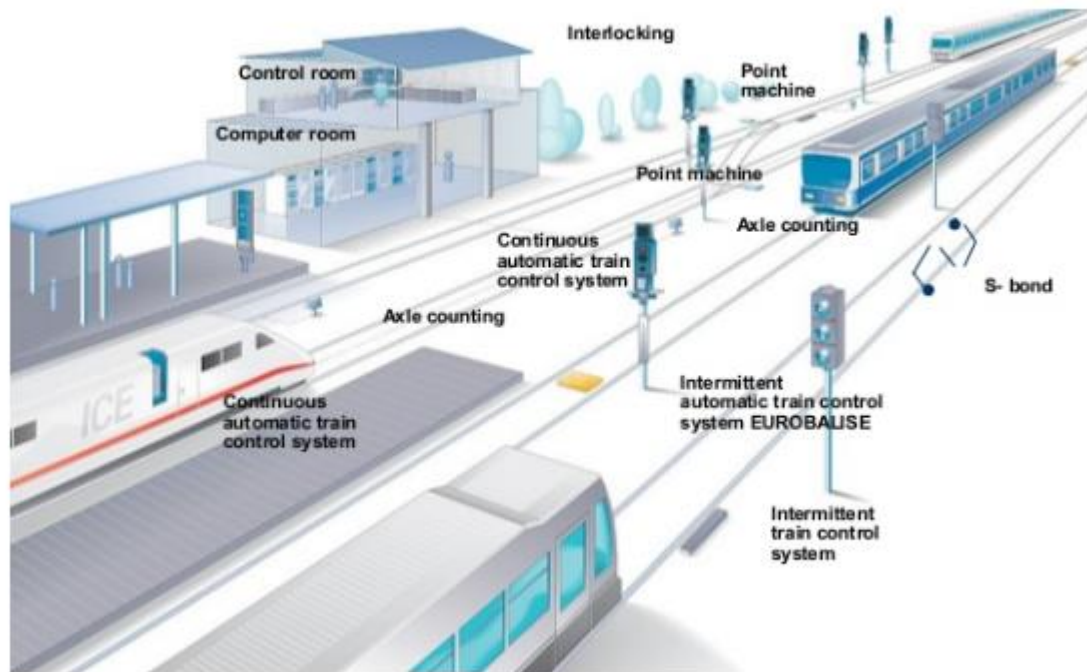


Figure 5 - Signal system overview.

- **Power Systems:** Metro stations rely on a stable and reliable power supply to operate trains, lighting, ventilation, and other critical systems. The power systems at this layer include substations, transformers, switchgear, and backup power sources, such as generators and uninterruptible power supplies (UPS).

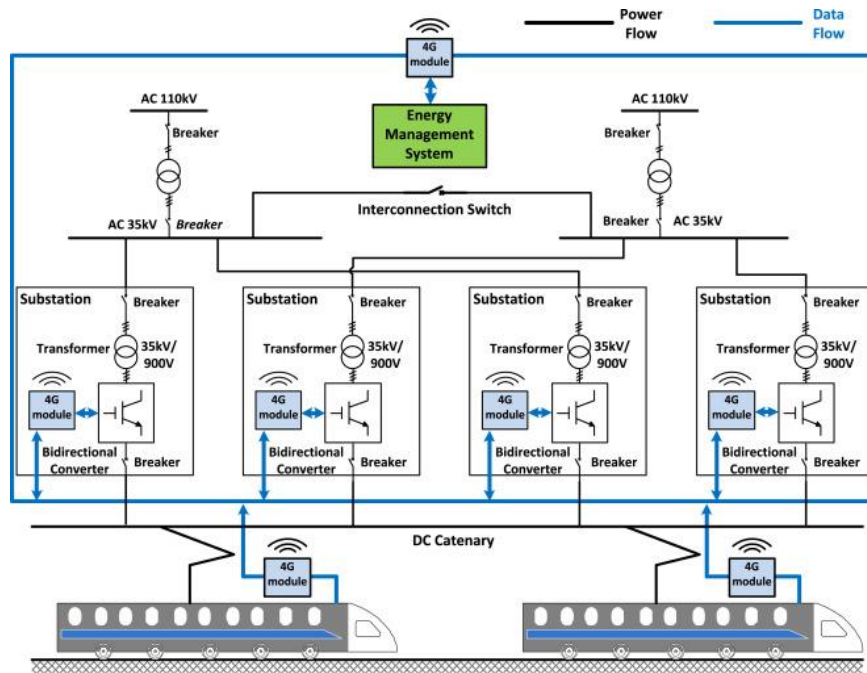


Figure 6 - Power system overview.

- **Environmental Control Systems:** These systems are responsible for maintaining a safe and comfortable environment within the metro station, including temperature, humidity, and air quality control. They may also include fire detection and suppression systems, as well as flood monitoring and control systems.
- **Security Systems:** Security systems, such as access control, surveillance cameras, and intrusion detection, are critical components of the OT layer in a metro station. While these systems enhance physical security, their increasing connectivity and digitization also introduce new vulnerabilities that attackers can exploit. Proper security measures, updates, and monitoring are essential to prevent unauthorized access, tampering, or disruption of these systems.

The vulnerabilities inherent in CBTC systems were starkly highlighted in November 2017 when the Domestic Security Council and the Cyber Council of the Intelligence and National Security Alliance (INSA) organized a tabletop exercise simulating a cyber-attack on critical infrastructure, with a power company as the target. Through this exercise, INSA and its members identified vulnerabilities within the system and outlined improvements needed for detection, response, and situational awareness (Ir Henry CHEUNG, 2019). Such exercises underscored the importance of proactive cybersecurity measures in safeguarding critical infrastructure from cyber threats.

Railways are not immune to cyber-attacks, as demonstrated by several incidents targeting railway systems worldwide. The OT layer presents a unique set of risks and vulnerabilities that can be exploited by malicious actors to disrupt metro station operations, compromise passenger safety, and cause significant damage. Some of the key risks at this layer include:

1. **Legacy Systems:** Many of the OT systems in metro stations are legacy systems that were designed and implemented before the advent of modern cybersecurity threats. These systems may have limited or no built-in security features, making them

vulnerable to cyberattacks. Attackers may exploit known vulnerabilities in these systems to gain unauthorized access, manipulate control logic, or inject malicious code.

2. **Proprietary Protocols:** OT systems often use proprietary communication protocols that are not well-understood or widely supported by mainstream cybersecurity tools and techniques. This can make it difficult to monitor these systems for signs of malicious activity or to implement effective security controls. Attackers may take advantage of these obscure protocols to evade detection and maintain persistence within the OT network.
3. **Air-Gapped Networks:** In some cases, OT systems may be isolated from the IT network and the internet through air-gapping, which involves physically separating the networks to prevent unauthorized access. However, air-gapping is not foolproof, and attackers may still be able to breach these networks through insider threats, social engineering, or the introduction of infected removable media.
4. **Lack of Security Awareness:** OT personnel, such as engineers and maintenance staff, may not have the same level of cybersecurity awareness as IT personnel. This can lead to risky behaviors, such as the use of default passwords, the connection of unauthorized devices to the OT network, or the failure to report suspicious activity. Attackers may exploit this lack of awareness to gain initial access to the OT network and move laterally to more critical systems.
5. **Supply Chain Risks:** The OT layer relies on a complex supply chain of hardware, software, and services from multiple vendors. Each of these components presents a potential entry point for attackers, who may exploit vulnerabilities in the supply chain to introduce malicious code, backdoors, or other threats into the OT environment. Metro station operators may have limited visibility into the security practices of their suppliers, making it difficult to assess and mitigate these risks.

Let's consider a specific attack vector involving a vulnerability in a legacy train control system component. In this scenario, the metro station uses an older, proprietary train control system that has been in place for several decades. While the system has undergone some updates and maintenance over the years, it still relies on outdated software and hardware components that are no longer supported by the original vendor.

One of these legacy components is a communication module responsible for transmitting critical data between the train control system and the individual trains. This module uses a proprietary protocol that was developed in-house by the vendor and lacks modern security features, such as strong encryption and authentication mechanisms.

An attacker, who has spent months researching the metro station's infrastructure and gathering intelligence on its systems, discovers this vulnerability through a combination of techniques, such as:

1. Obtaining leaked documentation or insider information about the train control system's architecture and components.
2. Analyzing publicly available information, such as job postings or procurement records, that mention the specific vendor or technology used in the system.
3. Conducting reconnaissance on the metro station's network, such as port scanning or traffic analysis, to identify potential entry points and weaknesses.

Armed with this knowledge, the attacker develops a custom exploit that targets the vulnerable communication module. The exploit takes advantage of a buffer overflow vulnerability in the module's firmware, allowing the attacker to remotely execute arbitrary code on the device.

To deliver the exploit, the attacker sets up a rogue wireless access point near the metro station, disguising it as a legitimate device that is part of the train control system's infrastructure. When the vulnerable communication module connects to this access point, the attacker's malicious code is injected into the module's memory, granting the attacker unauthorized access to the train control system.

With this foothold in the OT layer, the attacker can now proceed to manipulate the train control system's functionality, potentially causing severe disruptions and safety hazards. The specific actions the attacker takes will depend on their objectives and the extent of control they have gained over the system.

The impact of a successful cyber-attack on a metro station's train control system can be severe and far-reaching, affecting the station's operations, safety, and reputation. Let's explore these impacts in more detail:

Operations:

- **Service disruptions:** The attack on the train control system can lead to significant service disruptions, causing trains to stop, delay, or operate at reduced speeds. This can result in a cascading effect throughout the metro network, leading to widespread delays and cancellations.
- **Financial losses:** The metro station may incur significant financial losses due to the disruption of services, as well as the costs associated with emergency response, system repairs, and investigations. Additionally, the station may face penalties or fines from regulatory bodies for failing to maintain a secure and reliable service.
- **Reduced capacity:** The attack may force the metro station to operate at a reduced capacity for an extended period, as authorities work to assess the damage, restore systems, and implement additional security measures. This can lead to overcrowding, longer wait times, and increased customer dissatisfaction.

Safety:

- **Collisions and derailments:** In the worst-case scenario, the attacker's manipulation of the train control system could cause trains to collide or derail, resulting in significant property damage, injuries, or even loss of life. This can occur if the attacker overwrites safety-critical functions, such as speed limits, braking systems, or train separation protocols.
- **Passenger panic and chaos:** The sudden and unexpected disruption of train services, combined with a lack of clear information and guidance, can lead to passenger panic and chaos. This can result in stampedes, fights, or other dangerous behaviours as people attempt to evacuate trains or stations, leading to further injuries and damage.
- **Fire and toxic hazards:** In some cases, the attack on the train control system may cause electrical or mechanical malfunctions that can lead to fires or the release of toxic substances.

This can pose a significant risk to passengers, staff, and emergency responders, as well as cause long-term damage to the station's infrastructure and environment.

Reputation:

- **Loss of public trust:** A major cyber-attack on a metro station can severely erode public trust in the station's ability to provide safe, reliable, and secure transportation services. Passengers may fear for their safety and choose alternative modes of transport, leading to a long-term decline in ridership and revenue.
- **Media scrutiny and negative publicity:** The attack will likely attract significant media attention, both locally and internationally. The metro station may face intense scrutiny and criticism over its cybersecurity practices, crisis response, and overall management. This negative publicity can damage the station's brand, making it harder to attract new customers, investors, or partners.
- **Political and regulatory consequences:** The attack may prompt investigations and inquiries from political leaders, regulators, and law enforcement agencies. The metro station may face legal and financial consequences if it is found to have been negligent or non-compliant with relevant cybersecurity standards and regulations.

In May 2022, a notable example occurred when a ransomware attack disrupted railway station gates in Germany, affecting the control systems responsible for managing access to station platforms. The malware encrypted critical files and databases associated with access control systems, rendering them inaccessible and leading to disruptions in passenger flow. Such incidents underscore the evolving threat landscape faced by railway operators and the potential consequences of cyber-attacks on railway infrastructure.

The power system plays a crucial role in metro stations by providing the electrical energy necessary for various components and facilities to function. However, an attack on the power station serving a metro facility can have significant consequences, like those mentioned above, including disruptions and safety concerns. Moreover, the proposed system's dependency on the power grid for electricity supply introduces additional vulnerabilities. As demonstrated by the August 2019 power cut in the United Kingdom, major outages can have cascading effects on transportation, public services, and the economy (Hockaday, 2019). In the context of the proposed system, a power grid outage would directly impact train operations, including signalling systems and station facilities, exacerbating the consequences of a cyber-attack.

Furthermore, the Human Layer is often considered one of the most vulnerable aspects of IT/OT systems due to the potential for human error, negligence, or malicious intent to compromise security (Kour, 2020). Attackers exploit human psychology using techniques like phishing, pretexting, or baiting to gain unauthorized access to systems or information. The Ukrainian power grid attack serves as a prominent example highlighting these vulnerabilities, where attackers initially compromised the utilities' IT networks and subsequently targeted Operational Technology (OT) systems responsible for controlling and monitoring physical processes, such as power distribution. Exploiting vulnerabilities in these OT systems allowed the attackers to gain control over industrial control devices (CyberLaw, 2015). Therefore, ensuring robust cybersecurity measures across all layers of the system, including the human layer, is essential to mitigate potential risks and safeguard critical infrastructure.

The proposed system, with its reliance on CBTC technology and interconnected control systems, would be vulnerable to similar cyber-attacks. Furthermore, the functionality of the operational control centre's OT devices could be compromised, impairing its ability to effectively manage and monitor train movements. Without accurate and reliable data from the control centre's OT devices, operators may struggle to coordinate train schedules, allocate resources efficiently, and respond to emergencies promptly. This could exacerbate disruptions and prolong the recovery process following a cyber-attack.

To mitigate these risks, metro station operators must adopt a multi-layered approach to security that addresses the unique challenges of the OT environment. This may include:

1. **Asset Discovery and Inventory:** Conducting a thorough inventory of all OT assets, including hardware, software, and communication protocols, to identify potential vulnerabilities and prioritize security efforts.
2. **Network Segmentation:** Implementing strong network segmentation between IT and OT networks, as well as between different OT systems, to limit the potential impact of a breach and prevent lateral movement.
3. **Access Control:** Enforcing strict access controls for OT personnel, including multi-factor authentication, role-based access, and least privilege principles, to minimize the risk of insider threats and unauthorized access.
4. **Patch and Vulnerability Management:** Establishing a regular patch and vulnerability management program for OT systems, prioritizing critical security updates and mitigating known vulnerabilities in a timely manner.
5. **Continuous Monitoring:** Deploying specialized OT security monitoring solutions, such as industrial intrusion detection systems (IDS) and anomaly detection tools, to detect and respond to potential threats in real-time.
6. **Incident Response Planning:** Developing and testing incident response plans that are specific to the OT environment, including procedures for isolating affected systems, communicating with stakeholders, and restoring normal operations.
7. **Supply Chain Risk Management:** Implementing a robust supply chain risk management program, including vendor risk assessments, security requirements in contracts, and ongoing monitoring of supplier security practices.
8. **Training and Awareness:** Providing regular cybersecurity training and awareness programs for OT personnel, including hands-on simulations and exercises, to promote a culture of security and encourage the reporting of suspicious activity.

Here's a detailed table addressing specific mitigations for each layer affected by the attack.

Table 1 - Mitigations for each layer

Layer	Mitigation Measures
OT Layer	Regularly update and patch systems. Implement network segmentation. Monitor for anomalous activity. Conduct vulnerability assessments and penetration testing. Implement strong authentication mechanisms
IT Layer	Enforce strong access controls. Encrypt sensitive data. Conduct regular security audits, Implement security monitoring and incident response tools. Deploy advanced endpoint protection solutions
Communication Layer	Implement secure communication protocols. Use firewalls to filter traffic. Monitor for unauthorized access attempts. Conduct network security assessments. Implement secure remote access solutions
Physical Layer	Enhance access control systems. Implement video surveillance. Develop contingency plans for emergency situations. Conduct physical security assessments. Provide physical security awareness training
Human Layer	Provide cybersecurity awareness training. Establish clear incident response procedures. Encourage a culture of security awareness. Provide specialized training for key personnel. Regularly review and update security policies

By implementing these measures and continually adapting to the evolving threat landscape, metro station operators can significantly reduce the risks associated with the OT layer and ensure the safety, reliability, and resilience of their critical infrastructure.

4. Conclusion

The operational technology (OT) layer in metro stations forms a critical component of their infrastructure, encompassing systems and devices essential for safe and efficient operations. However, this layer is not immune to cybersecurity risks, as demonstrated by the vulnerabilities inherent in legacy systems, proprietary protocols, air-gapped networks, lack of security awareness, and supply chain risks. The scenario of a cyber-attack targeting a legacy train control system component illustrates the potential severity of such threats, with possible impacts ranging from service disruptions and financial losses to safety hazards and damage to reputation. Moreover, the interconnected nature of modern metro systems, coupled with their reliance on power systems and communication technologies, further amplifies the potential consequences of cyber-attacks.

Cyber attackers have evolved from isolated individuals to organized groups of digital criminals targeting critical infrastructure. In the United Kingdom, for example, 90% of key installations have experienced at least one successful cyber-attack, highlighting the pervasive nature of this threat (Ir Henry CHEUNG, 2019). If such attacks were to occur on railway systems, the consequences could be severe, including disruptions to mobility, city functionality, and public safety. The interconnectedness of railway systems with urban infrastructure underscores the importance of proactive cybersecurity measures to mitigate the impact of cyber-attacks and ensure the resilience of railway operations.

There were approximately 200 CBTC installations worldwide by the end of 2018, many of which utilize Wi-Fi connections operating on publicly accessible 2.4GHz or 5.8GHz bands for communication with trains. Additionally, an Ethernet-based connection is established between computers and systems, with an 'connect as necessary' internet connection. This highlights the existence of access paths that hackers could potentially exploit to infiltrate railway infrastructure, posing a vulnerability that could impact railway safety. Andrew Ginter, vice president of industrial security for Waterfall Security, explains, 'A large fraction of attacks that caused OT consequences did so by compromising machines in the IT network exclusively'

(Nelson, 2024) Therefore, it is imperative for railway operators to prioritize cybersecurity measures to safeguard against such threats and ensure the resilience and continuity of railway operations. Moreover, collaboration between stakeholders, including government agencies, regulatory bodies, industry partners, and cybersecurity experts, is essential to effectively address the evolving threat landscape and ensure the long-term security of metro systems worldwide.

In conclusion, the increasing reliance on interconnected systems and the growing sophistication of cyber threats underscore the urgent need for a comprehensive and proactive approach to cybersecurity in the railway sector. By implementing robust security measures, fostering collaboration among stakeholders, and continuously adapting to the changing threat landscape, metro system operators can enhance the resilience of their infrastructure and protect the millions of passengers who rely on their services every day.

Failure to address cybersecurity risks adequately in the railway sector can lead to severe consequences, including widespread service disruptions, financial losses, compromised passenger safety, and damage to the reputation of railway operators. Without robust security measures, railway systems are vulnerable to cyber-attacks, which can have cascading impacts on both metro stations and broader urban infrastructure, eroding public trust. Inadequate cybersecurity measures not only expose sensitive passenger data to theft or manipulation but also exacerbate the potential harm caused by cyber incidents. Therefore, it is crucial for railway operators to prioritize cybersecurity to mitigate these risks and ensure the continued safety, reliability, and resilience of metro systems.

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